Arunabha Majumder

Mechanical R&D Engineer | Mechanisms, Prototyping and Testing

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SUMMARY

Mechanical engineer who likes to understand how things work and build better versions. For my PhD at Aalborg University I designed a shoulder exoskeleton with a new spring and motor mechanism, built and tested many prototypes, and made the test rigs myself. As a student I built a Go-Kart, a hybrid vehicle and a rehabilitation exoskeleton. I sketch ideas by hand before CAD.

EXPERIENCE

PhD Researcher, Mechatronics and Robotics (project VIEXO) | 2023 - Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Designed a hybrid shoulder exoskeleton where a spring carries most of the arm load and a small 6 Nm motor adds the rest; it cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Designed the mechanism and parts in SolidWorks and iterated them with 3D printing, fixing parts that broke or flexed too much until they held up in testing.
- Built and maintained my own test rigs and integrated the motor, encoder and force sensors with the frame.
- Validated a force estimate on a 5-bar parallel robot against a reference force sensor, within 10% RMS error.
- Collected data from 12 participants in user tests; worked in a team of 3 on the assistance controller (my part: equations such as the Fourier fit, and GUI behaviour). Taught and supervised students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic artificial muscles to an existing lab design, collected test data on the lab's setup and used it to design the outer PID loop of a gain-scheduled controller (0.3-0.78% error, IEEE CONECCT 2022).

Student Builds, Siddaganga Institute of Technology | 2017 - 2019
Tumakuru, India

- Designed, machined, assembled and user-tested an upper-limb rehabilitation exoskeleton (Best Major Project Award, INR 100,000 TATA Technologies grant).
- Built a Go-Kart (2017) and a human-electric hybrid vehicle for Efficycle (2018) with student teams.

SKILLS

- **Design:** SolidWorks (CSWA), mechanisms and moving assemblies, hand sketching
- **Prototyping:** 3D printing, test rigs; machining and assembly (Bachelor's)
- **Test and data:** Motors, encoders, force/torque and pressure sensors, DAQ, Arduino; MATLAB, Python
- **Drawing:** Diploma in Fine Arts; concept sketches and technical illustrations
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Denmark

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, India

BTech, Mechanical Engineering | 2015 - 2019

Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer. Also IEEE CONECCT 2022.

REFERENCES

Available upon request.